MagicCode Super-Resolution Software Release Notes

Document Version: v2.0.0. Release Date: September 2, 2026.

1. Summary

MagicCode Super-Resolution Software v2.0.0 publishes four public alg_mode_e values (spatial Speed/Balanced and temporal Speed/Balanced) and documents the session-API contract for temporal super-resolution on Android Vulkan/OpenGLES, Apple Metal, and Windows Vulkan / Direct3D 11 / OpenGL 4.3.

2. Key Changes

- Public version string is v2.0.0 (MC_GetVersion).
- alg_mode_e: SPATIAL_SPEED_MODE=0, SPATIAL_BALANCED_MODE=1, TEMPORAL_SPEED_MODE=2, TEMPORAL_BALANCED_MODE=3.
- Temporal scaler range is (1, 8]. Spatial remains [1.0, 8.0] subject to backend and model.
- Formal temporal entry is MC_Init / MC_Process / MC_Uninit with magic_frame_t + temporal_frame_t (color/output, depth, motion, jitter, frame_index/reset_history, camera, gpu_context, command-buffer).
- MC_Enable still processes one color texture and is not a complete temporal entry.
- Windows product archives: lib/windows/libmagic_sr.lib and libmagic_enable_sr.lib. Enable does not accept MAGIC_BACKEND_D3D11.
- Both temporal modes default to deriving a missing reactive mask (and approximate transparency on the Metal/Vulkan/GLES Speed path) and to depth+MV history rejection. Explicit masks win per channel.

3. Compatibility

Platform	GPU backends	Temporal
Android	Vulkan, OpenGLES	Temporal GPU yes; CPU temporal no
iOS / iPadOS / macOS	Metal	Temporal GPU yes; CPU temporal no
Windows	Vulkan, Direct3D 11, OpenGL 4.3	Temporal GPU yes; CPU temporal no

- Operating systems: Windows 10 64-bit and above, Android 8.0 and above, iOS 13.0 and above, iPadOS 13.0 and above, and macOS 12.0 and above.
- Input resolution: 64×64 to 4032×4032.

- CPU SIMD (spatial CPU paths): AVX2 on x86_64, NEON on ARM64. CPU temporal is unavailable.

Temporal Super-Resolution

v2.0.0 adds temporal super-resolution through the session API: libmagic_sr.a (Windows: libmagic_sr.lib) + mc_interface.h + MC_Init / MC_Process / MC_Uninit. MC_Enable remains a single-texture spatial helper. It may accept a temporal alg_mode_e value by range check, but it cannot supply per-frame depth, motion, or jitter and must not be documented as a complete temporal entry.

Modes

Value	Enumerator	Role
0	SPATIAL_SPEED_MODE	Spatial, throughput first
1	SPATIAL_BALANCED_MODE	Spatial, quality/cost trade-off
2	TEMPORAL_SPEED_MODE	Temporal, throughput first
3	TEMPORAL_BALANCED_MODE	Temporal, canonical FSR-family path

- Temporal scaler_factor is (1, 8]; 1.0 itself is rejected. Spatial remains [1.0, 8.0].
- TEMPORAL_SPEED_MODE on Metal, Vulkan, and OpenGLES currently uses a lightweight Convert → Activate → Upscale pipeline. Direct3D 11 and desktop OpenGL 4.3 Speed use the FSR-family temporal path (same family as Balanced, not a separate customer API).
- TEMPORAL_BALANCED_MODE uses the canonical FSR-family path on every GPU temporal backend. Desktop Balanced edge-resolve is supported on Windows Vulkan, Windows Direct3D 11, desktop OpenGL 4.3, and Apple Metal. Android Vulkan and Android OpenGLES do not run edge-resolve. If enable_sharpening is on, RCAS runs after edge-resolve.
- CPU backends (X86 / NEON) cannot run temporal modes.
- The API accepts (1, 8]. Exact 2× has optimized kernels. Not every scale or device is claimed as device-run.
- Internal pipeline names are implementation detail, not public configuration.

Minimum per-frame inputs

- magic_frame_t.image_in / image_out: caller-owned color and output GPU resources.
- magic_frame_t.frame: non-NULL temporal_frame_t* with struct_size = sizeof(temporal_frame_t).
- temporal_frame_t.depth and motion at input resolution.
- jitter_offset_x/y in input pixels, top-left, +X right, +Y down.
- frame_index / reset_history; camera_near / camera_far / camera_fov_y (radians).
- gpu_context at MC_Init: Vulkan requires physical_device + device; D3D11 requires device; Metal device may be NULL; GL/GLES use the current context (library never makes current).
- Vulkan temporal requires magic_frame_t.command_buffer to be a recording VkCommandBuffer. Metal command_buffer is optional. D3D11 / GL / GLES leave it NULL.
- Create-stage depth_reversed / depth_infinite / hdr_color on input_param_t (all-zero = conventional-Z, finite far, LDR). Immutable for the handle lifetime.

Motion, depth, and masks

- Motion is current → previous. The library never negates MVs.
- motion_vector_scale_x/y is the only MV numeric conversion. (0,0) defaults to (input_width, input_height). Mixed zero is rejected.
- Depth is GPU device depth in [0, 1]. Linear view-space depth is not accepted.
- reactive / transparency are optional. Explicit caller textures win per channel.
- When omitted, the library derives an approximate mask. That is not a semantic transparency mask.
- Speed on Metal/Vulkan/GLES may derive both missing reactive and an approximate transparency. FSR-family paths (Balanced on all backends; Speed on D3D11/desktop GL) derive missing reactive only and bind an internal zero transparency texture.
- Both temporal modes enable depth+MV history rejection by default.
- Diagnostic only: MAGICSR_TAAU_AUTO_MASK=0 and MAGICSR_TAAU_HIST_REJECT=0 can turn those defaults off. They are not a customer configuration path.

4. Migration Notes

- From v1.2.x: keep numeric 0/1 for spatial modes (now named SPATIAL_*). Add 2/3 only when integrating temporal.
- Replace any MC_Process(handle, image_in, image_out) usage with MC_Process(handle, magic_frame_t*).
- Set input_param_t.struct_size and temporal_frame_t.struct_size.
- Do not treat Enable temporal enums as a full temporal integration.
- Review Privacy Policy and Data Reporting Notice before distributing applications that include the SDK or plugins. Those legal documents are unchanged by this technical release.

5. Known Limitations

- Not every backend, scale, or device class is device-run. Distinguish API support from verified configurations.
- Derived masks are approximations. Omitting transparency is not equivalent to a semantic transparency mask.
- Vulkan temporal requires a recording command buffer and explicit layouts; the library does not begin, end, or submit it.
- Performance depends on device, driver, backend, resolution, scale, and motion quality.

6. Validation Snapshot

API contracts are defined by interface/mc_interface.h and the GPU host implementations. Compile/static coverage includes several temporal rates; that is not a device-run of every rate. Plugin Enable smoke remains spatial.